Command Line Interface Calculator

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Introduction

This will be our first official project we work on together. Reason for choosing this as our first project is that many of the programming and the maths required for making this project a success has already been taught to us by our university. So, use this project to revise your knowledge and most importantly learn new things. Unlike med students or Business students we can practice what we learn from day one. So, take this opportunity to show off your skills and passion.

CLI Calculator is a calculator that can be used to calculate wide variety of calculations. Like,

- Basic Operations
- Area
- Volume
- Sequences
- Physics

And lot more…

What is Command line Interface (CLI)?

Since this is our first project, we will be making it a Command line interface program which means the user needs to interact with the console and type in inputs. Instead of a Graphical User Interface (GUI) which would make it pretty and easy for the user, but for now CLI works.

What you will need for this project:

- Python
- Python IDE (Thonny is fine but there are lot better IDEs out there like Pycharm)
- Github Desktop
- Discord for PC (optional)

Project Structure

This project will have two main files.

- main.py
- functions.py

main.py will have all the code required for taking in inputs and outing outputs. It will be responsible for communicating/Interacting with the user. Error handling is a must! The main loop of the program starts here.

- Displaying the menus.
- Taking inputs from the user.
- Calling the appropriate functions from functions.py.
- Displaying the results to the user.
- Handling errors and invalid inputs.
- Controlling the main loop of the program.

functions.py will have all the maths and the calculation code of the program. Its role is to receive inputted data from the main.py and return the answers back to main.py.

- Receive the required data from main.py.
- Perform the required calculation.
- Return the calculated result back to main.py.

more details about each of these files will be given in the future.

I also plan to add jobs.xlsx and readme.md file. jobs will have a UpToDate record of the work that needs to be done and are already completed. Readme will have a small description about the project.

Functions

This will be a new programming concept for some members of the team. Which is why I added this as a main topic. For the ones who will be working on functions.py here is a small description of what a function is and how it will be used in this project.

Click here to view the documentation

Now that you have some understanding of what a function is let's make the addition function,

The structure of python function is,

def function_name(parameters):

    return answer

here is a practical example:

creation of the addition function

```
def addition(num_1, num_2):
  total = num_1 + num_2
  return total
```

#use of addition function

```
value = addition(10,15)

print(value)
```

Output: -

25

We will be making such functions for all the other calculations as well, and they will be in the functions.py. **Remember jobs.xslx will have what parameters it will take and what it should return as the answer.** So, make sure when you are working on a function to read it first.

Here are our Initial starting functions,

Basic Operators

- Addition
- Subtraction
- Multiplication
- Division
- Modulus

Area

- Square
- Rectangle
- Circle
- Triangle
- Other required shapes

Volume

- Cube
- Cuboid
- Sphere
- Cylinder
- Other required solids

Sequences

- Arithmetic sequence
- Geometric sequence

Physics

- Velocity
- Projectiles

Details about these functions will be available in jobs.xslx

Assignments

The assignments below are the **initial areas of responsibility** for each member.

This is not a restriction, anyone here can work in any file, but I want you to work on these assigned files first before working on the other one. The assignments are based on the current individual Python experience of the team members.

main.py	Thisew, Kishani,Anosh
functions.py	Shivantha, Anosh, Nethusha

Everyone is encouraged to communicate with the other members when their work depends on another part of the project.

Since main.py and functions.py need to work together, any changes to function names, parameters, or return values should be communicated to the rest of the team. Lot of this will be covered when we start using Github.

Final expected results

this project should give us a good understanding of collaborative software development and coding, while producing a functional calculator that we can continue expanding with more calculations and features.

The main expected outcomes are:

- A fully working command-line calculator.

- A good understanding of Python functions.
- A better understanding of mathematical programming and formula implementation.
- Experience working with multiple Python files.
- Experience working as a team on a shared codebase.
- Experience using GitHub and version control.
- Better understanding of writing and maintaining code written by other team members.
- Improved communication and collaboration skills.
- Experience planning, assigning, and tracking programming tasks.